

Text-Centric Visual Question Answering Under Visual Degradation: A Comparative Study of Modular and End-to-End Approaches

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Abstract—Text-centric Visual Question Answering (VQA) answers natural language queries by reading and reasoning over text in visual content. In real-world settings, visual degradation introduces uncertainty in optical character recognition (OCR) and downstream reasoning. While both modular OCR-based pipelines and end-to-end vision-language models are used for text-centric VQA, their comparative behavior under degraded visual conditions remains underexplored. This paper presents an empirical comparison using 4013 images with 7000 question-answer pairs, evaluated through accuracy, similarity metrics, OCR error measures, and inference throughput. Results show fine-tuned modular pipelines achieve up to 57.50% exact-match accuracy versus 38.00% for end-to-end baselines in strict correctness. The modular pipelines employ SA-DBNet, a custom text detection architecture combining a ResNet-18 backbone with self-attention spatial modeling and deformable convolutions in a Feature Pyramid Network. Critically, our analysis reveals that conventional OCR error metrics (CER, WER) do not have a straightforward relationship with downstream VQA performance, as semantic reasoning can compensate for certain recognition failures when contextual cues are present.

Index Terms—Text-centric VQA, OCR, Vision-Language Models, Degraded Visual Conditions

I. INTRODUCTION

Visual Question Answering (VQA) seeks to respond to natural language inquiries on image content by concurrently analyzing visual and textual data. In this larger job, “text-centric VQA” is a very difficult setting since answering a question needs correctly reading and understanding the text in the image. In real life, these kinds of situations happen when reading product labels, deciphering signs, getting numbers from printed materials, and analyzing language in papers or natural sceneries. In the actual world, writing in pictures is sometimes hard to see because of motion blur, low resolution, compression errors, or busy backgrounds. These issues often result in flawed optical character recognition (OCR), yielding noisy or partial textual inputs for later reasoning. A lot of the research that has already been done believes that higher OCR accuracy will automatically lead to better VQA performance. Recent advancements in extensive multimodal models indicate that semantic thinking may, in certain instances, offset recognition failures, especially when contextual cues are present. This insight prompts a more thorough investigation into the impact of OCR errors resulting from visual degradation on subsequent VQA performance.

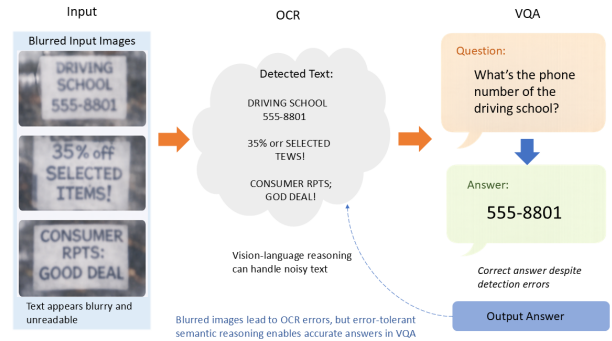


Fig. 1. Conceptual illustration of text-centric VQA from a drone perspective. The system must recognize and reason over text (e.g., labeling) under motion blur degradation.

In this study, we examine this issue through an empirical analysis of modular OCR-assisted pipelines and end-to-end vision-language models for text-centric visual question answering under compromised visual conditions. We create a dataset with 4013 images and 7000 question-answer pairs that includes a variety of text types and layouts. A text detection module, a text recognition module, and a language reasoning component make up two modular pipelines. They are tested against an end-to-end vision-language model. All systems are evaluated using a standardized process that includes tests for exact correctness, strong similarity, semantic alignment, and inference efficiency. The resulting comparison gives a clear picture of how different system designs work when the visuals go worse and how the quality of OCR affects how well the system answers questions later on. The investigation elucidates the degree to which semantic reasoning might alleviate suboptimal text recognition in text-centric VQA contexts. This work has three important parts. Initially, we present a benchmark framework for text-centric Visual Question Answering (VQA) in compromised visual environments. Second, we conduct a controlled empirical comparison of modular OCR-assisted pipelines, their pretrained equivalents, and an end-to-end vision-language model, utilizing both stringent and robustness-focused assessment criteria with runtime analysis. Third, we present an empirical analysis of the correlation between OCR accuracy and ultimate VQA performance, emphasizing the

significance of downstream reasoning in text-oriented VQA systems.

II. RELATED WORK

Text-centric visual question answering (VQA) has been introduced to address scenarios where answering a question requires reading and reasoning over textual content embedded in images. Early benchmarks such as TextVQA [1] and ST-VQA [2] established the necessity of integrating optical character recognition (OCR) into VQA pipelines, demonstrating that conventional object-centric models fail when answers depend on scene text. Subsequent datasets, including OCR-VQA and DocVQA [3], extended this paradigm to document images and structured layouts, further reinforcing the role of text extraction as a core component of visual reasoning. In parallel, significant progress has been made in OCR itself, with robust text detectors such as CRAFT [7] and DBNet [9], and recognizers such as TrOCR [6] and SVTR, enabling more accurate transcription of irregular and degraded text. Many text-centric VQA systems explicitly leverage these advances by feeding recognized text tokens into multimodal reasoning models, often assuming that improvements in OCR quality directly translate to improved VQA performance. More recently, end-to-end vision-language models such as CLIP, Flamingo, BLIP-2, and Qwen-VL [8] have sought to bypass explicit OCR by jointly learning visual perception and semantic reasoning, suggesting that high-level context may compensate for imperfect text recognition. However, despite these developments, there is limited systematic analysis of how OCR errors under realistic visual degradation propagate to downstream VQA performance, or whether traditional OCR metrics such as CER and WER are reliable indicators of final answer correctness. Existing works primarily report aggregate VQA accuracy under standard conditions, leaving the relationship between low-level transcription noise and high-level semantic reasoning largely unexplored. This gap motivates our empirical study, which directly benchmarks modular OCR-assisted pipelines against an end-to-end vision-language model under controlled visual degradation to analyze how OCR quality influences text-centric VQA outcomes.

III. METHODOLOGY

Text-centric visual question answering (VQA) requires jointly interpreting visual content and embedded text to answer natural language queries. In practical acquisition scenarios, however, textual cues are often observed under non-ideal visual conditions, including motion blur, defocus, and resolution loss. To explicitly capture this setting, let $I \in \mathbb{R}^{H \times W \times 3}$ denote an image and q a corresponding question. The observed input is modeled as a degraded version of the original image, $\tilde{I} = \mathcal{D}(I)$, where $\mathcal{D}(\cdot)$ represents a visual degradation operator. The objective of text-centric VQA is therefore to predict an answer a conditioned on the degraded visual input and the question:

$$a = f(\tilde{I}, q). \quad (1)$$

This formulation makes explicit that all perception and reasoning stages must operate on imperfect visual evidence, which is central to the empirical analysis conducted in this work.

A common paradigm for addressing text-centric VQA decomposes the mapping $f(\cdot)$ into modular stages corresponding to text detection, text recognition, and semantic reasoning. Under this formulation, the degraded image $\tilde{I}$ is first processed by a text detection function $r(\cdot)$ to produce a set of text regions:

$$B = r(\tilde{I}), \quad (2)$$

where $B = \{b_1, b_2, \dots, b_K\}$ denotes the detected text bounding boxes. These regions are subsequently converted into textual tokens through a recognition function $h(\cdot)$:

$$T = h(B), \quad (3)$$

yielding a sequence of recognized tokens $T = \{t_1, t_2, \dots, t_M\}$. Finally, a semantic reasoning module $g(\cdot)$ integrates the recognized text with the question to generate the predicted answer:

$$a = g(T, q). \quad (4)$$

This modular decomposition enables the effects of OCR quality and semantic reasoning to be examined independently, allowing a controlled analysis of how recognition errors propagate to downstream VQA performance.

In contrast, end-to-end vision-language models directly infer answers from visual and linguistic inputs without explicitly separating OCR as an intermediate step:

$$a = f_{\text{VLM}}(\tilde{I}, q), \quad (5)$$

where $f_{\text{VLM}}(\cdot)$ denotes a pretrained vision-language model. Such models implicitly learn joint visual-semantic representations and may therefore mitigate error amplification caused by explicit OCR failures. Including this formulation enables a direct comparison between modular and end-to-end approaches under identical degradation conditions.

To evaluate performance comprehensively, we employ metrics that capture both strict correctness and robustness to recognition noise. Exact-match accuracy measures the proportion of predictions that exactly match the ground-truth answers and is defined as:

$$\text{Acc} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(\hat{a}_i = a_i), \quad (6)$$

where N denotes the total number of question-answer pairs. To account for partial correctness arising from minor transcription errors, we additionally report Average Normalized Levenshtein Similarity (ANLS):

$$\text{ANLS} = \frac{1}{N} \sum_{i=1}^N \max \left(0, 1 - \frac{D(a_i, \hat{a}_i)}{\max(|a_i|, |\hat{a}_i|)} \right), \quad (7)$$

where $D(\cdot, \cdot)$ denotes the Levenshtein distance. Semantic similarity is measured using embedding-based similarity to capture meaning-level agreement beyond surface-form matching.

Character Error Rate (CER) and Word Error Rate (WER) are reported as diagnostic indicators of OCR quality, while Frames Per Second (FPS) quantifies end-to-end inference throughput.

Two modular OCR-assisted pipelines are evaluated, differing in their detection and recognition components while sharing a common semantic reasoning backbone. Both pipelines employ Qwen2-VL-2B as the reasoning module, ensuring that observed differences in VQA performance are primarily attributable to OCR quality rather than reasoning capacity.

A. SA-DBNet: Self-Attention Enhanced Text Detection

The text detection component in the first modular pipeline is based on SA-DBNet, a purpose-built architecture designed for robust text localization under visual degradation. SA-DBNet extends the Differentiable Binarization (DB) framework [9] with two key modifications.

First, a self-attention spatial bottleneck is injected at the deepest backbone feature level. Operating on the $1/32$ -scale feature map from a ResNet-18 [12] backbone, the module computes query, key, and value projections through 1×1 convolutions, forming an attention map over all spatial positions. A learnable scaling parameter γ controls the residual contribution of the attended features. This global spatial reasoning allows the detector to leverage long-range contextual dependencies, which is particularly valuable when text regions are small, sparse, or surrounded by visually similar clutter—conditions typical of degraded aerial imagery.

Second, the Feature Pyramid Network employs deformable convolutions (DCN-v2) [13] at every lateral output level. Unlike standard convolutions with fixed grid sampling, deformable convolutions learn data-dependent spatial offsets, enabling the receptive field to adaptively warp around geometrically distorted text. This is essential for handling perspective-skewed and curved text common in real-world captures.

SA-DBNet produces three output maps: a probability map P , a threshold map T , and a differentiable binary map $B = 1/(1 + e^{-k(P-T)})$ with $k = 50$. The model is trained using a composite loss combining Binary Cross-Entropy for the probability map, L1 loss for the threshold map, and Dice loss for the binary map. Training is performed using AdamW with gradient clipping. The model is first trained on TextVQA pseudo-labels (21,953 images) and then fine-tuned on ICDAR 2015 (1000 images) at 640×640 resolution, achieving an F1-score of 59% on the detection benchmark.

The first pipeline integrates SA-DBNet for text detection with transformer-based recognition. Text recognition is carried out using TrOCR [6], which formulates OCR as a sequence-to-sequence translation task using a Vision Transformer encoder and an autoregressive language decoder.

The second modular pipeline emphasizes inference efficiency and practical deployability. It is based on PP-OCrv4, which employs DB++ with a MobileNetV3 backbone for text detection, coupled with a DBFPN neck and a DBHead for differentiable binarization. Text recognition is performed using SVTR, which balances representational capacity and computational efficiency through spatial mixing operations.

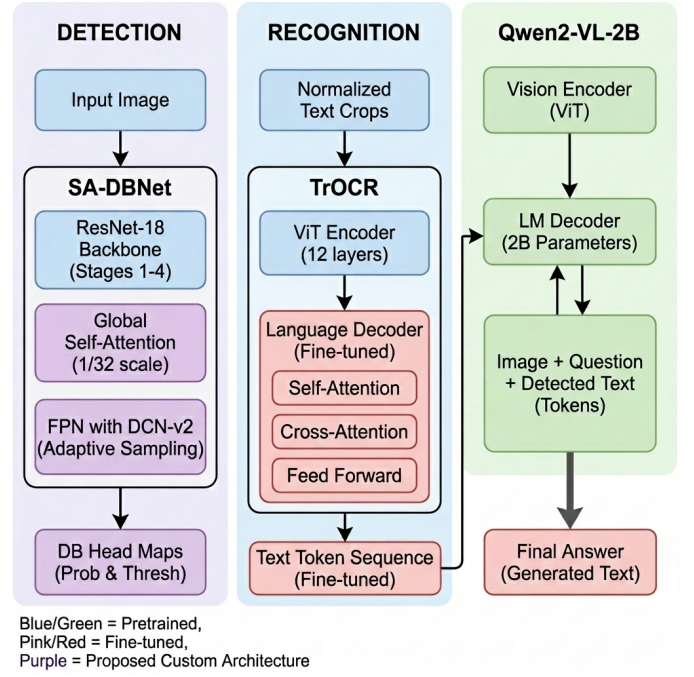


Fig. 2. Proposed modular pipeline architecture: SA-DBNet + TrOCR + Qwen2-VL. Color coding: blue = pretrained components, purple = proposed custom architecture (self-attention bottleneck, DCN-v2 FPN, DB head), pink/red = fine-tuned components, green = pretrained reasoning module. The ResNet-18 backbone uses pretrained ImageNet weights; all other SA-DBNet components are trained from scratch.

Both modular pipelines share the same reasoning component. During inference, Qwen2-VL-2B operates with 4-bit quantization and Flash Attention 2 to reduce memory usage and improve throughput. To enhance robustness under severe visual degradation, targeted fine-tuning strategies are applied to the OCR components. PaddleOCR is trained using a three-stage curriculum learning scheme with progressively increasing blur severity. TrOCR is fine-tuned on blur-augmented data starting from pretrained `trocr-base-stage1` weights, incorporating Gaussian blur, motion blur, and resolution degradation during training while keeping the Vision Transformer encoder frozen.

IV. EXPERIMENTS AND RESULTS

This section presents a detailed empirical evaluation of modular OCR-assisted pipelines and an end-to-end vision-language model for text-centric VQA under degraded visual conditions. We first describe the benchmark and experimental protocol, followed by a comparative performance analysis, an assessment of domain-specific fine-tuning, and a focused examination of OCR quality metrics in relation to downstream VQA accuracy.

A. Dataset and Experimental Setup

The evaluation is conducted on a curated benchmark consisting of 4013 images associated with a total of 7000 question-answer pairs. The dataset is designed to reflect realistic text-centric VQA scenarios, encompassing a wide

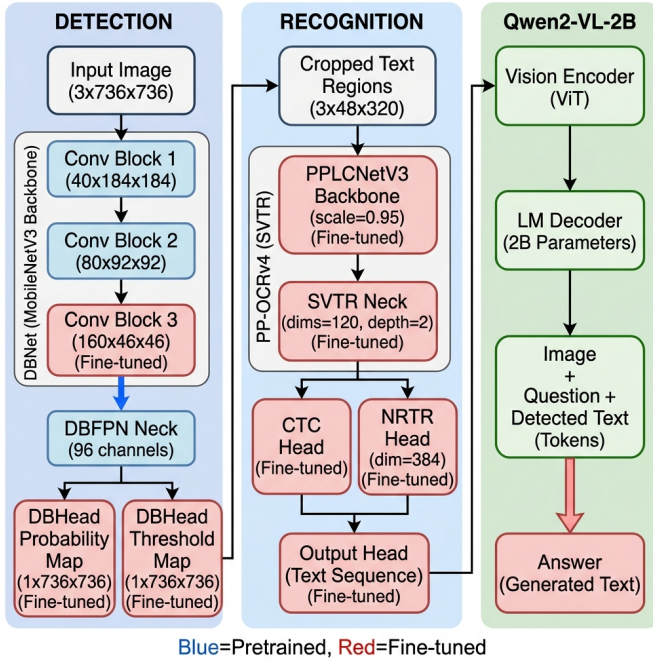


Fig. 3. Baseline PaddleOCR + Qwen pipeline architecture showing the standard DBNet+SVTR modular approach for benchmark comparison.

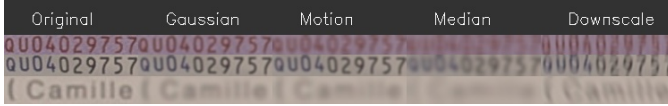


Fig. 4. Examples of visual degradation applied to text images in the experimental setup, including Gaussian blur, motion blur, median filtering, and downscaling.

range of textual content, visual layouts, and scene contexts. Images are drawn from text-heavy real-world environments and exhibit mixed viewpoints, background clutter, and non-uniform illumination, all of which pose challenges for both text extraction and semantic reasoning.

To simulate acquisition artifacts commonly encountered in mobile and aerial imaging systems, visual degradation is applied uniformly across the dataset using Gaussian blur with standard deviation $\sigma = 5.0$. This controlled degradation setting captures the effects of motion blur and defocus while ensuring consistency across experimental conditions. In addition to Gaussian blur, we consider representative examples of motion blur, median filtering, and downscaling to characterize the diversity of degradation effects applied during dataset preparation, as illustrated in Fig. 4.

Questions are categorized into three groups based on their reasoning requirements: (i) header-level queries that depend on prominent textual cues, (ii) detail-level queries requiring accurate recognition of fine-grained or numerical text, and (iii) spatial reasoning queries involving the relative arrangement of textual elements. This categorization enables a comprehensive evaluation of OCR fidelity and semantic reasoning under degraded visual input.

TABLE I
MAIN PERFORMANCE COMPARISON (BLUR $\sigma = 5.0$, $n = 7000$)

Pipeline	Acc (%)	ANLS	Semantic	CER	FPS
PaddleOCR+Qwen (FT)	42.83	0.59	0.61	0.42	0.72
SA-DBNet+TrOCR+Qwen (FT)	57.50	0.54	0.50	0.46	0.59
Qwen2-VL (End-to-End)	38.00	0.57	0.65	0.58	0.76

B. Implementation Details

For modular pipelines, text detection and recognition are implemented using established open-source frameworks. The SA-DBNet-based pipeline employs the custom SA-DBNet detector for text localization and a fine-tuned TrOCR model for recognition, while the PaddleOCR-based pipeline uses PP-OCRv4 with curriculum-trained weights optimized for blur robustness. In both cases, recognized text tokens are provided to Qwen2-VL-2B through a unified prompting interface, ensuring that differences in performance arise primarily from OCR quality rather than variations in reasoning input.

The end-to-end baseline directly applies the Qwen2-VL model to the degraded image and question without explicit OCR decomposition. All experiments are conducted under identical hardware and inference settings. End-to-end throughput is reported in Frames Per Second (FPS), accounting for the complete inference pipeline, including preprocessing, OCR (when applicable), and semantic reasoning.

C. Comparative Performance Analysis

Table I presents the primary quantitative comparison under heavy visual degradation ($\sigma = 5.0$). Both fine-tuned modular pipelines demonstrate superior exact-match accuracy compared to the end-to-end baseline. The SA-DBNet+TrOCR pipeline achieves the highest accuracy at 57.50%, followed by PaddleOCR at 42.83%. In comparison, the end-to-end Qwen2-VL model attains 38.00%. This validates that explicit text extraction provides stronger grounding for VQA when inputs are degraded.

However, the end-to-end model achieves comparable throughput at 0.76 FPS versus 0.72 FPS for PaddleOCR+Qwen, a marginal 5.6% difference while SA-DBNet+TrOCR operates at 0.59 FPS due to its more heavy-weight recognition stage.

D. Impact of Domain-Specific Fine-Tuning

To isolate the effect of OCR adaptation, we compare fine-tuned pipelines against their pretrained counterparts under identical blur conditions. Results are summarized in Table II. Fine-tuning yields substantial improvements for both pipelines. PaddleOCR improves from 32.50% to 42.83%, while SA-DBNet+TrOCR improves from 28.00% to 57.50%, resulting in a remarkable 29.50 point gain.

These results confirm that domain-specific exposure to degraded visual inputs is critical for robust text recognition in text-centric VQA. The larger relative gains observed for SA-DBNet+TrOCR further highlight the adaptability of

TABLE II
IMPACT OF FINE-TUNING ON BLUR ROBUSTNESS

Pipeline	Pretrained	Fine-tuned	Gain
PaddleOCR+Qwen	32.50%	42.83%	+10.33%
SA-DBNet+TrOCR+Qwen	28.00%	57.50%	+29.50%

TABLE III
OCR QUALITY METRICS (FINE-TUNED MODELS)

Pipeline	CER ↓	WER ↓	Semantic ↑
PaddleOCR (FT)	0.42	0.50	0.61
SA-DBNet+TrOCR (FT)	0.46	0.56	0.50

transformer-based recognition models when trained under challenging visual conditions.

E. OCR Quality and Downstream VQA Performance

Table III reports OCR-specific quality metrics. PaddleOCR achieves lower Character Error Rates (0.42) compared to SA-DBNet+TrOCR (0.46) but scores significantly lower in downstream VQA Accuracy (42.83% vs 57.50%). This inversion suggests that lower aggregate CER does not always guarantee better VQA performance; the nature of the errors (e.g., missing critical keywords vs minor typos) plays a larger role in semantic reasoning success.

V. ABLATION STUDY

To better understand the factors influencing text-centric VQA performance under visual degradation, we conducted a set of ablation experiments that differentiate the effects of individual OCR integration, domain-specific fine-tuning, and OCR quality on downstream reasoning. All ablations are performed under identical blur conditions ($\sigma = 5.0$).

A. Effect of Explicit OCR Integration

We first examine the impact of individual OCR by comparing modular OCR-assisted pipelines with an end-to-end vision-language model (Qwen2-VL). As shown in Table IV, including standalone OCR consistently improves the accuracy of the exact-match (+6.4%), using the specialized recognition capabilities of the modular components. However, the End-to-End model offers competitive throughput (0.76 FPS) and higher semantic similarity.

TABLE IV
COMPARISON OF MODULAR VS. END-TO-END APPROACHES (BLUR $\sigma = 5.0$)

Pipeline Design	Accuracy	Sem. Sim.	FPS
End-to-End (Qwen2-VL)	38.00%	0.65	0.76
Modular (PaddleOCR+Qwen)	42.83%	0.61	0.72
Modular (SA-DBNet+TrOCR+Qwen)	57.50%	0.50	0.59

B. Importance of Text Detection

To assess the necessity of a dedicated text detector, we evaluated the recognition performance with and without the detection stage (feeding the full scene image directly to the recognizer). Furthermore, isolated testing of SA-DBNet revealed extremely robust bounding box detection capabilities, achieving a high F1-score of 59% on its own prior to any text recognition. Table V reveals a substantial performance drop when detection is removed (ANLS 0.54 $\rightarrow$ 0.09), confirming that even powerful recognizers like TrOCR require localized feature support in complex, cluttered scenes.

TABLE V
ABLATION OF TEXT DETECTION MODULE (ANLS)

Configuration	ANLS Score
Full Pipeline (Det + Rec)	0.54
No Detection (Rec Only)	0.0886

C. Relationship Between OCR Quality and VQA Accuracy

Finally, we highlight a key finding: OCR quality metrics (CER/WER) are not linear predictors of VQA success. While the End-to-End model has a higher theoretical ‘‘OCR Error’’ (as it does not output transcripts), its semantic reasoning allows it to answer questions correctly 38% of the time, often not requiring perfect character-level recognition. This shows the need for task-specific evaluation metrics like ANLS over pure recognition metrics in downstream applications.

VI. DISCUSSION

The experimental results provide useful information on the behavior of text-centric VQA systems under degraded visual conditions. Across all evaluations, the domain-specific adaptation of OCR components consistently improves downstream performance, indicating that aligning training data with deployment conditions is beneficial for robust text-centric VQA. In particular, curriculum-based fine-tuning and blur-aware training enable OCR models to better handle challenging visual input without compromising their general recognition capability. The comparison between modular OCR-assisted pipelines and the end-to-end vision-language model highlights complementary strengths. Modular pipelines achieve higher exact-match accuracy under severe degradation, reflecting the advantage of explicit textual grounding when precise recognition is required. At the same time, the end-to-end model offers higher inference throughput and competitive robustness on similarity-based metrics, making it well suited for efficiency-oriented scenarios. These observations suggest that both paradigms are effective and that the choice between them can be guided by practical application requirements.

In addition, the results indicate that the OCR quality and the final VQA performance are not strictly coupled. Even when recognition errors are present, semantic reasoning modules are often able to recover correct answers by using contextual information. This behavior suggests that the nature of

recognition errors—whether they affect answer-critical tokens such as proper nouns and numerical values, or peripheral contextual text—is a more reliable predictor of VQA success than aggregate error rate. Future evaluation frameworks should therefore incorporate task-aware error weighting rather than relying solely on character-level metrics. This behavior emphasizes the importance of evaluating text-centric VQA systems as integrated pipelines rather than optimizing individual components in isolation.

In general, the findings reinforce the value of systematic, system-level evaluation for text-centric VQA and demonstrate that modular and end-to-end approaches can perform reliably under realistic visual degradation when appropriately designed.

VII. CONCLUSION

This paper presented an empirical comparison of modular OCR-assisted pipelines and an end-to-end vision-language model for text-centric visual question answering under degraded visual conditions. The results reveal clear trade-offs between the two paradigms we have evaluated. Fine-tuned modular pipelines achieve higher exact match accuracy by providing explicit textual grounding, while the end-to-end vision-language model offers competitive robustness on similarity based metrics together with higher inference throughput. These observations indicate that adapting OCR components to target degradation conditions is beneficial when strict textual correctness is required. The analysis further shows that conventional OCR error measures, such as Character Error Rate (CER) and Word Error Rate (WER), do not consistently reflect downstream VQA performance. Even under severe visual degradation scenarios, semantic reasoning modules are often able to recover correct answers by leveraging contextual and linguistic cues. This highlights the importance of evaluating text-centric VQA systems holistically, rather than relying solely on OCR-centric metrics.

Future work will extend the evaluation to additional degradation types including motion blur and defocus, expand the dataset to strengthen statistical confidence, and explore lightweight hybrid approaches that integrate explicit OCR with end-to-end vision-language reasoning.

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